

SolarLab Technical and User Manual

Technical guide to thin-film solar-cell simulation, workflows, and validation

SolarLab Project

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Chapter 1

Executive Summary

SolarLab is a thin-film photovoltaic device simulator that couples drift-diffusion transport, Poisson electrostatics, mobile-ion redistribution, recombination, and optical generation within a reproducible software workflow. The numerical core is exposed through a Python API, a FastAPI service layer, and a TypeScript/Plotly workstation. An experimental two-dimensional extension supports lateral microstructure and grain-boundary studies.

This manual serves two purposes:

1. It introduces the device variables, parameter conventions, and simulation workflow for readers who are new to SolarLab or to semiconductor device simulation.
2. It documents the implemented physics, numerical method, validation gates, assumptions, and limitations needed for technical review and reproducible use.

The manual is based on the current SolarLab repository state:

- Commit: 43c81d7fef009ffca598f58f85a40ad4e661e1e
- Validation date: 2026-05-19
- Primary simulator tree: `perovskite-sim/`

The validation evidence collected before writing this PDF is summarized in Chapter 16. The short version is:

Validation gate	Result
<code>pytest</code>	647 passed, 1 skipped
<code>pytest -m slow</code>	72 passed
<code>pytest -m validation</code>	18 passed
<code>npm run build</code>	passed
<code>npm run test:run</code>	320 passed

The validation evidence should be interpreted as implementation evidence, not as a guarantee of predictive accuracy for arbitrary material stacks. It shows that the equations, numerical coupling, backend/frontend interfaces, regression envelopes, and documented benchmark cases are internally consistent for the repository state listed above.

Chapter 2

How To Read This Manual

If you are new to device simulation, read Chapters 2 to 6 in order. These chapters explain the device, the variables, the equations, and the numerical method before introducing the software workflow.

If you are already familiar with semiconductor drift-diffusion, start with Chapters 5, 7, 8, and 11:

- Chapter 5 defines the model equations.
- Chapter 7 explains the solver.
- Chapter 8 explains YAML/Python/API workflows.
- Chapter 11 gives validation evidence and limitations.

The notation in this manual follows standard solar-cell convention:

- V_{oc} : open-circuit voltage.
- J_{sc} : short-circuit current density.
- FF : fill factor.
- PCE : power-conversion efficiency.
- n : electron density.
- p : hole density.
- P : mobile positive ion-vacancy density.
- ϕ : electrostatic potential.
- E : electric field.
- G : optical generation rate.
- R : recombination rate.

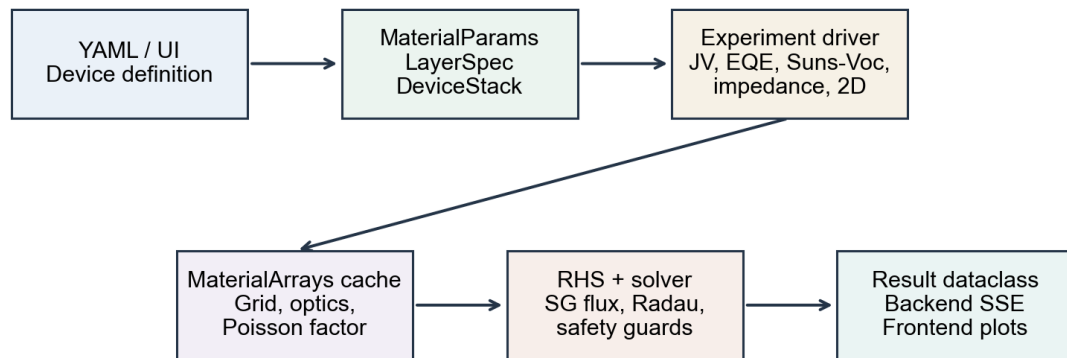
All simulator inputs are SI unless explicitly stated otherwise. Length is in meters, time in seconds, current density in $A m^{-2}$, density in m^{-3} , mobility in $m^2 V^{-1} s^{-1}$, and diffusion coefficient in $m^2 s^{-1}$. The electron affinity χ and band gap E_g are stored in eV.

Chapter 3

Simulator Architecture

SolarLab should be read as a coupled model implementation rather than as a stand-alone user interface. YAML files, the Python API, the FastAPI backend, and the TypeScript frontend all operate on the same device schema. A parameter edited in the browser is therefore serialized into the same scientific object used by the solver.

A



Scientific inputs, cached material arrays, numerical solving, and user-facing results remain separate.

Figure 3.1: SolarLab architecture flow

The architecture preserves four technical boundaries:

- device definition is separated from experiment settings;
- material and grid arrays are cached before the time integrator runs;
- physics hooks are activated by the resolved simulation mode and by the presence of required parameters;
- backend and frontend layers transport result dataclasses instead of reimplementing solver logic.

For reproducible interpretation, each reported curve should identify the device stack, physics tier, ex-

periment driver, and solver configuration. Missing metadata weakens reproducibility even when the numerical run completes successfully.

Chapter 4

What SolarLab Simulates

SolarLab represents a solar cell as a sequence of material layers between two contacts. The default coordinate follows the stack direction. Incident light generates electron-hole pairs, while the electrostatic field and contact selectivity drive carrier separation and extraction. In perovskite devices, mobile ionic defects can redistribute under the same field, producing history-dependent current-voltage response.

The default solver is one-dimensional and is appropriate for:

- J-V sweeps and hysteresis;
- dark diode curves;
- EQE and electroluminescence;
- impedance;
- transient photovoltage;
- ion-coupled degradation;
- tandem current matching;
- screening many material candidates.

SolarLab also includes an experimental two-dimensional solver. The 2D solver extrudes the same vertical device stack laterally and is intended for:

- 1D/2D parity checks;
- vertical grain boundaries;
- lateral microstructure;
- $V_{oc}(L_g)$ grain-size sweeps.

The 2D solver is not a replacement for the 1D workflow. Mobile ions are held as a static Poisson background during 2D J-V runs; this is an explicit model assumption and should be reported when interpreting 2D results.

Device Structure (n-i-p Perovskite Solar Cell)

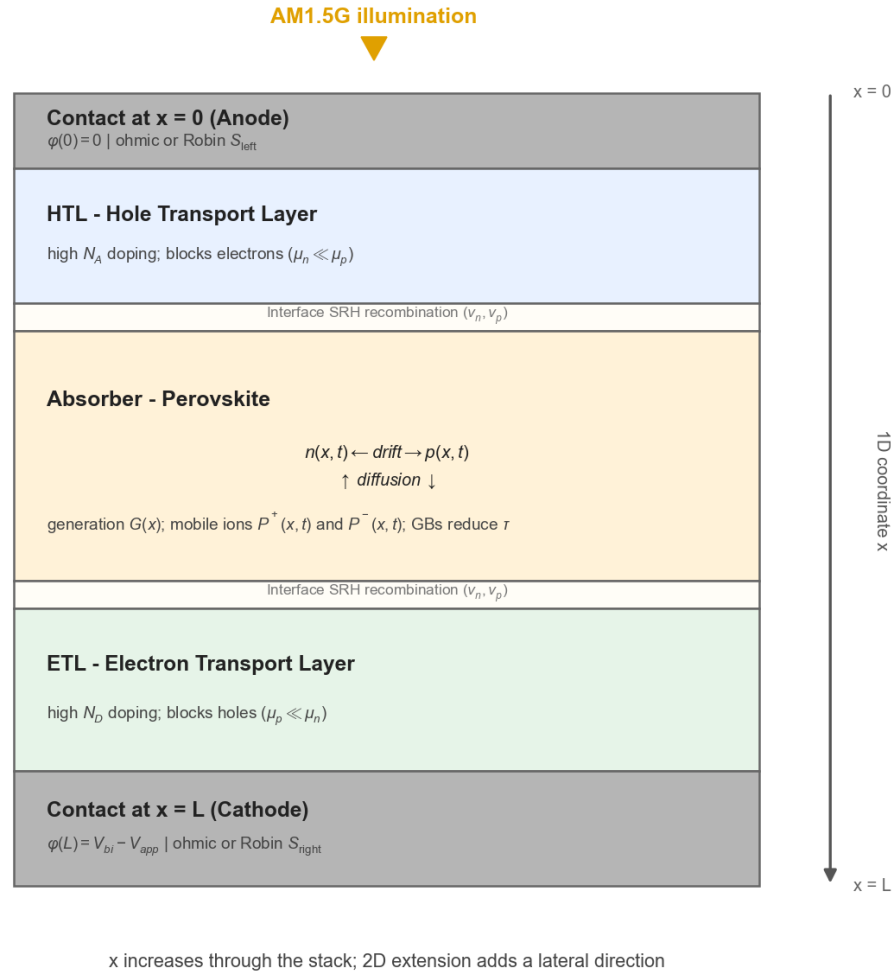


Figure 4.1: SolarLab device structure

Energy Band Diagram

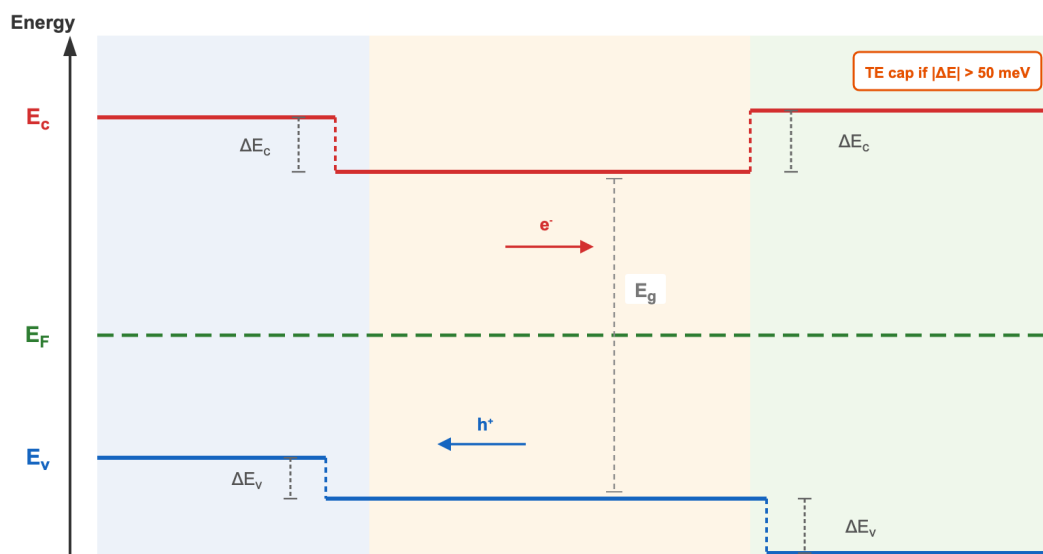


Figure 4.2: SolarLab band-diagram concept

Chapter 5

Beginner Physics Primer

5.1 Electrons, Holes, And Bands

In a semiconductor, electrons can move in the conduction band and holes can move in the valence band. A solar cell works because light creates electron-hole pairs and the device structure makes it more likely that electrons leave through one contact while holes leave through the other.

SolarLab stores two energy parameters that are essential for heterojunctions:

- χ , the electron affinity;
- E_g , the band gap.

Together these determine approximate conduction-band and valence-band positions. At an interface, a change in χ or E_g creates band offsets. Band offsets are important because they can block one carrier type while allowing the other carrier type to pass.

5.2 Generation And Recombination

Optical generation $G(x)$ is the rate at which absorbed photons create electron-hole pairs per unit volume. Recombination $R(n, p)$ is the rate at which electrons and holes annihilate. A good solar cell has high generation and carrier collection, but low recombination.

SolarLab includes several recombination channels:

- Shockley-Read-Hall (SRH) trap-assisted recombination;
- radiative recombination;
- Auger recombination;
- interface recombination through surface recombination velocities;
- optional trap-density profiles near interfaces.

5.3 Mobile Ions And Hysteresis

Perovskite cells can contain mobile ionic defects. These defects move much more slowly than electronic carriers. During a voltage scan, the ion profile may not reach equilibrium at each voltage. This creates J-V hysteresis. SolarLab treats this as a physical transient, not as a post-processing correction.

In the state vector, the default mobile ion variable is P , the positive vacancy density. A second negative species can be configured through `D_ion_neg`, `P0_neg`, and `P_lim_neg`.

Chapter 6

Device Definition

6.1 Core Data Flow

The simulator data flow is:

YAML or inline device dictionary
-> MaterialParams + LayerSpec + DeviceStack
-> experiment driver
-> MaterialArrays cache
-> result dataclass

The main dataclasses are:

- MaterialParams: physical parameters for one material.
- LayerSpec: layer name, role, thickness, and material parameters.
- DeviceStack: full multilayer stack and global device settings.

6.2 Layer Roles

Layer roles used by the frontend and YAML schema are:

Role	Meaning
substrate	optical-only layer, excluded from electrical drift-diffusion
front_contact	front electrode or transparent conductor
ETL	electron transport layer
absorber	active photovoltaic absorber
HTL	hole transport layer
back_contact	rear electrode

Substrate layers must form a contiguous prefix of the stack. They participate in transfer-matrix optics but do not appear in the drift-diffusion grid.

6.3 Main Material Parameters

Field	Meaning	Unit
eps_r	relative permittivity	dimensionless
mu_n, mu_p	electron and hole mobilities	$m^2 V^{-1} s^{-1}$
D_ion	positive ion diffusion coefficient	$m^2 s^{-1}$
P_lim	positive ion steric limit	m^{-3}
P0	initial positive ion density	m^{-3}
ni	intrinsic carrier density	m^{-3}
tau_n, tau_p	SRH lifetimes	s
n1, p1	SRH trap reference densities	m^{-3}
B_rad	radiative recombination coefficient	$m^3 s^{-1}$
C_n, C_p	Auger coefficients	$m^6 s^{-1}$
alpha	Beer-Lambert absorption coefficient	m^{-1}
N_A, N_D	acceptor and donor densities	m^{-3}
chi	electron affinity	eV
Eg	band gap	eV
optical_material	key into data/nk/*.csv	string
incoherent	incoherent TMM layer flag	boolean

Additional optional parameters cover negative ions, temperature scaling, field-dependent mobility, trap profiles, and optical fallbacks.

6.4 Parameter Dictionary

This section is intentionally detailed because input specification is often the dominant source of error in device simulation. SolarLab cannot infer whether a parameter comes from measurement, literature, fitting, or exploratory screening. Each field should therefore be treated as a documented scientific assumption.

6.4.1 Electrical And Transport Fields

Field	Unit	Physical role	Beginner guidance
eps_r	1	relative dielectric constant in Poisson's equation	Larger values screen charge and reduce field gradients. Use layer-specific literature values rather than a single generic perovskite value.
mu_n	$m^2 V^{-1} s^{-1}$	low-field electron mobility	Controls electron extraction and transport resistance. Remember that $1 \text{ cm}^2 V^{-1} s^{-1} = 10^{-4} m^2 V^{-1} s^{-1}$.
mu_p	$m^2 V^{-1} s^{-1}$	low-field hole mobility	Same unit conversion convention as mu_n. Low mobility typically affects FF before producing a large change in V_{oc} .
N_A	m^{-3}	ionized acceptor density	Represents p-type doping. Values reported in cm^{-3} must be multiplied by 10^6 .

Field	Unit	Physical role	Beginner guidance
N_D	m^{-3}	ionized donor density	Represents n-type doping. Use compensated doping only when it is part of the intended physical model.
ni	m^{-3}	intrinsic carrier density	Appears in mass action and SRH terms. It should be consistent with the effective band gap and temperature when doing parameter studies.
chi	eV	electron affinity	Sets approximate conduction-band alignment. Inconsistent χ values can introduce artificial transport barriers.
Eg	eV	band gap	Sets band offsets and thermodynamic interpretation. Beer-Lambert optics do not automatically shift spectral absorption when Eg changes.

6.4.2 Ion And Hysteresis Fields

Field	Unit	Physical role	Beginner guidance
D_ion	$m^2 s^{-1}$	positive mobile-ion diffusion coefficient	Set to zero when mobile ions are excluded from the model. Nonzero values can produce scan-rate dependence.
P0	m^{-3}	initial positive ion density	Represents the equilibrium mobile-vacancy population before redistribution.
P_lim	m^{-3}	steric upper density for positive ions	Limits ion accumulation through a finite-site-density approximation. The value should be consistent with the assumed mobile-site density.
D_ion_neg	$m^2 s^{-1}$	negative mobile-ion diffusion coefficient	Enables a second mobile species. Leave zero for the default single-ion model.
P0_neg	m^{-3}	initial negative ion density	Use only when the negative species is part of the physical hypothesis.
P_lim_neg	m^{-3}	steric upper density for negative ions	Same interpretation as P_lim.
E_a_ion	eV	Arrhenius activation energy for ion diffusion	Used for temperature-dependent ion mobility. Fitted or literature-derived values should be reported with provenance.

6.4.3 Recombination And Trap Fields

Field	Unit	Physical role	Beginner guidance
tau_n	s	electron SRH lifetime	Shorter lifetime increases trap-assisted recombination. It is often an effective fitted parameter.
tau_p	s	hole SRH lifetime	Interpret with tau_n; asymmetry can model carrier-selective trap response.

Field	Unit	Physical role	Beginner guidance
n1	m^{-3}	SRH electron reference density	Encodes the effective trap-energy position and should be changed consistently with the trap model.
p1	m^{-3}	SRH hole reference density	Companion to n1.
B_rad	$m^3 s^{-1}$	radiative recombination coefficient	Central to radiative-limit and photon-recycling studies.
C_n	$m^6 s^{-1}$	electron Auger coefficient	Most relevant when high carrier densities are expected.
C_p	$m^6 s^{-1}$	hole Auger coefficient	Same caution as C_n.
trap_N_t_interface	m^{-3}	interface-near trap density	Activates spatial trap profiles when supplied with decay information.
trap_N_t_bulk	m^{-3}	bulk trap density	The asymptotic trap density away from the interface.
trap_decay_length	m	decay length or Gaussian width	Must be physically resolvable by the grid.
trap_profile_shape	string	exponential or gaussian trap profile	Select the profile shape that corresponds to the assumed defect distribution.

6.4.4 Optical Fields

Field	Unit	Physical role	Beginner guidance
alpha	m^{-1}	scalar Beer-Lambert absorption coefficient	Suitable for simplified optical studies. It is not wavelength-resolved.
optical_material	string	key into tabulated n, k data	Required for TMM, EQE, and EL. The key must exist in perovskite_sim/data/nk.
n_optical	1	constant refractive-index fallback	Useful for approximate optical calculations but not a substitute for measured n, k .
incoherent	boolean	thick-layer incoherent TMM treatment	Intended for the first thick substrate. The current TMM path does not support arbitrary incoherent layers mid-stack.

6.4.5 Temperature And Field-Dependent Mobility Fields

Field	Unit	Physical role	Beginner guidance
Nc300	m^{-3}	conduction-band density of states at 300 K	Optional temperature-scaling input. Use only when the temperature model is calibrated.
Nv300	m^{-3}	valence-band density of states at 300 K	Companion to Nc300.
mu_T_gamma	1	mobility power-law temperature exponent	Default keeps a common scattering-style trend. Document any changed value.
B_rad_T_gamma	1	radiative coefficient temperature exponent	Default zero preserves legacy behavior.
varshni_alpha	eV K ⁻¹	Varshni band-gap coefficient	Zero disables band-gap temperature shift.

Field	Unit	Physical role	Beginner guidance
varshni_beta	K	Varshni temperature parameter	Used with varshni_alpha.
v_sat_n	$m s^{-1}$	electron velocity-saturation limit	Zero disables Caughey-Thomas saturation for electrons.
v_sat_p	$m s^{-1}$	hole velocity-saturation limit	Zero disables Caughey-Thomas saturation for holes.
ct_beta_n	1	electron Caughey-Thomas exponent	Controls sharpness of velocity saturation.
ct_beta_p	1	hole Caughey-Thomas exponent	Same role for holes.
pf_gamma_n	$(V/m)^{-1/2}$	electron Poole-Frenkel mobility coefficient	Zero disables field-enhanced hopping for electrons.
pf_gamma_p	$(V/m)^{-1/2}$	hole Poole-Frenkel mobility coefficient	Zero disables field-enhanced hopping for holes.

6.4.6 Device-Level Fields

Field	Unit	Physical role	Beginner guidance
V_bi	V	built-in voltage in Poisson boundary condition	This is not always identical to the derived heterostack Fermi-level separation.
Phi	$m^{-2} s^{-1}$	incident photon flux for Beer-Lambert generation	For spectral experiments use TMM data rather than only changing Phi.
T	K	device temperature	Affects thermal voltage and enabled temperature-dependent hooks.
mode	string	legacy, fast, or full physics tier	The mode is a ceiling; hooks still need required parameters.
interfaces	$m s^{-1}$ pairs	interface recombination velocities	One pair per internal electrical interface, ordered from the front contact toward the rear contact.
S_n_left	$m s^{-1}$	left electron contact velocity	None gives ohmic default. 0 is blocking. Large values approach ohmic extraction.
S_p_left	$m s^{-1}$	left hole contact velocity	Same convention as S_n_left.
S_n_right	$m s^{-1}$	right electron contact velocity	Same convention as S_n_left.
S_p_right	$m s^{-1}$	right hole contact velocity	Same convention as S_n_left.
microstructure	object	2D grain-boundary specification	Used by 2D drivers; ignored by standard 1D paths.

6.5 Global Device Parameters

DeviceStack stores:

Field	Meaning
V_bi	built-in voltage used in the Poisson boundary condition
Phi	incident photon flux for Beer-Lambert generation
interfaces	interface recombination velocities (v_n, v_p)
T	device temperature in K
mode	legacy, fast, or full

Field	Meaning
S_n_left, S_p_left, S_n_right, S_p_right	selective-contact velocities
microstructure	optional 2D grain-boundary block

6.6 Built-In Potential

SolarLab has two related built-in potentials:

1. `stack.V_bi`: the value read from YAML and used in the Poisson boundary condition.
2. `stack.compute_V_bi()`: a derived effective built-in potential computed from Fermi-level differences across the electrical stack.

This distinction matters. Some legacy and benchmark configurations use a manual V_{bi} convention. Heterostacks with χ and E_g also have band-alignment information that can imply a different effective Fermi-level separation.

The Poisson boundary is:

$$\phi(0) = \phi_{\text{left}}, \quad \phi(L) = \phi_{\text{left}} + V_{bi} - V_{\text{app}}.$$

The effective built-in potential is used for defaults such as the automatic upper voltage in J-V sweeps.

6.7 YAML Example

```
device:
  V_bi: 1.1
  Phi: 2.5e21
  T: 300.0
  mode: full
  contacts:
    left:
      S_p: 1.0e3
      S_n: 1.0e-3
    right:
      S_n: 1.0e3
      S_p: 1.0e-3

layers:
- name: HTL
  role: HTL
  thickness: 2.0e-7
  eps_r: 3.0
  mu_n: 1.0e-8
  mu_p: 1.0e-6
  ni: 1.0e9
  N_A: 1.0e24
  N_D: 0.0
```

```
D_ion: 0.0
P_lim: 1.0e27
P0: 0.0
tau_n: 1.0e-6
tau_p: 1.0e-6
n1: 1.0e9
p1: 1.0e9
B_rad: 0.0
C_n: 0.0
C_p: 0.0
alpha: 0.0
chi: 2.2
Eg: 3.0
```

YAML 1.1 can parse bare scientific notation such as `1e-9` as a string. SolarLab coerces numeric-looking strings to floats, but writing values as `1.0e-9` is clearer and safer.

Chapter 7

Governing Equations

This chapter states the governing equations in dimensional SI form. The implementation evaluates discretized node and face arrays, but the continuous form clarifies the model assumptions and the physical meaning of each state variable.

7.1 Poisson Equation

The electrostatic potential is obtained from:

$$\frac{\partial}{\partial x} \left(\varepsilon_0 \varepsilon_r \frac{\partial \phi}{\partial x} \right) = -\rho. \quad (7.1)$$

Here ρ includes electrons, holes, dopants, and mobile ions. The discretization uses a harmonic face permittivity:

$$\varepsilon_{i+1/2} = \frac{2\varepsilon_i \varepsilon_{i+1}}{\varepsilon_i + \varepsilon_{i+1}}. \quad (7.2)$$

The harmonic mean is appropriate for a sharp dielectric interface because it matches the series-capacitor limit.

Assumptions and limits:

- electrostatics is quasistatic;
- magnetic effects and wave propagation in the electrical solve are ignored;
- layer interfaces are abrupt on the scale of the grid;
- ρ is built from the configured mobile species, carriers, and dopants.

7.2 Carrier Continuity

The electron continuity equation is:

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} + G - R. \quad (7.3)$$

The hole continuity equation is:

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} + G - R. \quad (7.4)$$

In heterojunctions, SolarLab uses band-corrected potentials:

$$\phi_n = \phi + \chi, \quad \phi_p = \phi + \chi + E_g. \quad (7.5)$$

These effective potentials allow the same numerical flux formulation to represent conduction-band and valence-band offsets in heterojunction stacks.

Assumptions and limits:

- carriers are described by drift-diffusion transport, not ballistic transport;
- the model is most natural when local quasi-equilibrium is a reasonable approximation;
- strongly quantum-confined structures, tunneling-dominated contacts, and hot carrier effects require additional modeling beyond this implementation.

7.3 Scharfetter-Gummel Flux

The Bernoulli function is:

$$B(\xi) = \frac{\xi}{\exp(\xi) - 1}. \quad (7.6)$$

The electron face current is:

$$J_{n,i+1/2} = \frac{qD_n}{\Delta x} [B(\xi)n_{i+1} - B(-\xi)n_i]. \quad (7.7)$$

The hole face current is:

$$J_{p,i+1/2} = \frac{qD_p}{\Delta x} [B(\xi)p_i - B(-\xi)p_{i+1}]. \quad (7.8)$$

The Scharfetter-Gummel form is used because it preserves stable exponential fitting in drift-dominated regimes.

Assumptions and limits:

- the potential and material properties are represented between adjacent grid nodes by a face-based discretization;
- the method is robust for high drift fields, but accuracy still depends on resolving layer thicknesses, interfaces, and trap profiles.

7.4 Ion Migration

Positive ion vacancies satisfy:

$$\frac{\partial P}{\partial t} = -\frac{\partial F_P}{\partial x}. \quad (7.9)$$

Ions cannot leave the device, so:

$$F_P(0) = F_P(L) = 0. \quad (7.10)$$

The ion flux includes a steric saturation correction:

$$s(P) = \frac{1}{\max(1 - P_{\text{avg}}/P_{\text{lim}}, 10^{-6})}. \quad (7.11)$$

This term limits ion accumulation as the local density approaches the configured site-density limit.

Assumptions and limits:

- the default mobile species is a positive vacancy-like species;
- negative mobile species are optional and must be explicitly configured;
- contacts are ion-blocking in the implemented boundary condition;
- the 2D J-V solver treats ions as a frozen Poisson background.

7.5 Recombination

SRH recombination is:

$$R_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}. \quad (7.12)$$

Radiative recombination is:

$$R_{\text{rad}} = B_{\text{rad}}(np - n_i^2). \quad (7.13)$$

Auger recombination is:

$$R_{\text{Auger}} = (C_n n + C_p p)(np - n_i^2). \quad (7.14)$$

The total bulk rate is:

$$R = R_{\text{SRH}} + R_{\text{rad}} + R_{\text{Auger}}. \quad (7.15)$$

Interface recombination uses surface velocities and is converted to a local volumetric source term near the interface.

Assumptions and limits:

- SRH parameters are effective trap parameters unless trap-energy provenance is supplied;
- radiative and Auger coefficients should be layer-appropriate;
- interface recombination is localized numerically near the interface, so grid resolution matters when very large surface velocities are used.

7.6 Optical Generation

For Beer-Lambert absorption:

$$G(x) = \Phi\alpha(x) \exp \left[- \int_0^x \alpha(x') dx' \right]. \quad (7.16)$$

For transfer-matrix optics, SolarLab loads tabulated $n(\lambda)$ and $k(\lambda)$ data and computes a wavelength-resolved absorption profile. The generation profile is integrated over the incident spectrum and cached before the transient solve.

EQE and electroluminescence require this wavelength-resolved optical machinery; Beer-Lambert-only stacks do not contain sufficient spectral information for those experiments.

Assumptions and limits:

- Beer-Lambert optics are fast and useful for trends, but cannot represent interference, reflection, or wavelength-dependent collection;
- TMM quality is limited by the provenance and wavelength coverage of the supplied optical constants;
- synthetic or placeholder optical constants should be treated as workflow demonstrations, not material-specific evidence.

Chapter 8

Boundary And Initial Conditions

8.1 Potential

The potential is Dirichlet at both contacts:

$$\phi(0) = \phi_{\text{left}}, \quad \phi(L) = \phi_{\text{left}} + V_{\text{bi}} - V_{\text{app}}.$$

Forward bias reduces the built-in field.

8.2 Carrier Contacts

By default, carrier densities at contacts are ohmic pins derived from local charge neutrality and mass action:

$$np = n_i^2, \quad n - p = N_D - N_A.$$

The stable two-branch solution avoids numerical cancellation for heavily doped layers.

In `full` mode, selected contacts can use a Robin flux:

$$J = \pm qS(u - u_{\text{eq}}),$$

where u is n or p , and S is a surface recombination/extraction velocity. $S = 0$ is blocking; large S approaches the ohmic limit.

8.3 Ion Boundaries

Both positive and negative mobile ions use zero-flux boundaries. This reflects the physical assumption that ions redistribute inside the device but do not leave through the contacts.

8.4 Initial States

The dark equilibrium state starts from charge neutrality and mass action. The illuminated state is obtained by integrating the transient equations under illumination at the starting voltage.

For J-V, impedance, and degradation, this light-soaked initial state matters because ion and carrier memory affect the measurement.

Chapter 9

Physics Tiers

SolarLab uses three fidelity modes:

Mode	Active physics	Typical use
legacy	disables upgraded hooks	historical benchmark reproduction
fast	build-once upgrades, no expensive per-RHS hooks	screening
full	every configured hook	high-fidelity single runs

The tier is a ceiling, not a command to fabricate missing data. For example:

- TMM requires `optical_material`.
- Field-dependent mobility requires nonzero `v_sat_*` or `pf_gamma_*`.
- Selective contacts require finite `S_*` values.
- Radiative reabsorption requires TMM/photon-recycling support.

Chapter 10

Numerical Method

The solver discretizes Eqs. 7.1–7.16 on the device grid and then advances the coupled state in time. The important point for users is that a J-V point is not a closed-form diode equation; it is the terminal current read from a relaxed drift-diffusion state at a specified applied voltage.

10.1 Grid

The 1D solver builds a multilayer grid over electrical layers. Substrate layers are excluded from the electrical grid but retained for TMM optics.

10.2 Method Of Lines

The PDEs are discretized in space first. The result is a coupled system of ordinary differential equations in time. This is the method-of-lines approach.

The state vector contains the node values for:

$$\mathbf{y} = (n, p, P)$$

and optionally a negative ion species.

10.3 Time Integration

SolarLab uses SciPy `solve_ivp`, primarily with the implicit Radau method. Selected difficult J-V steps use bounded fallback logic, including step bisection and BDF fallback. The code caps `max_step` around voltage transitions to avoid large near-flat-band integration jumps.

10.4 Poisson Fast Path

The Poisson equation becomes a tridiagonal linear system. SolarLab prefactors this operator and reuses the factorization. This is why `build_material_arrays` must be called outside the RHS and not inside the time-step loop.

Solver Pipeline: How Each Timestep Works

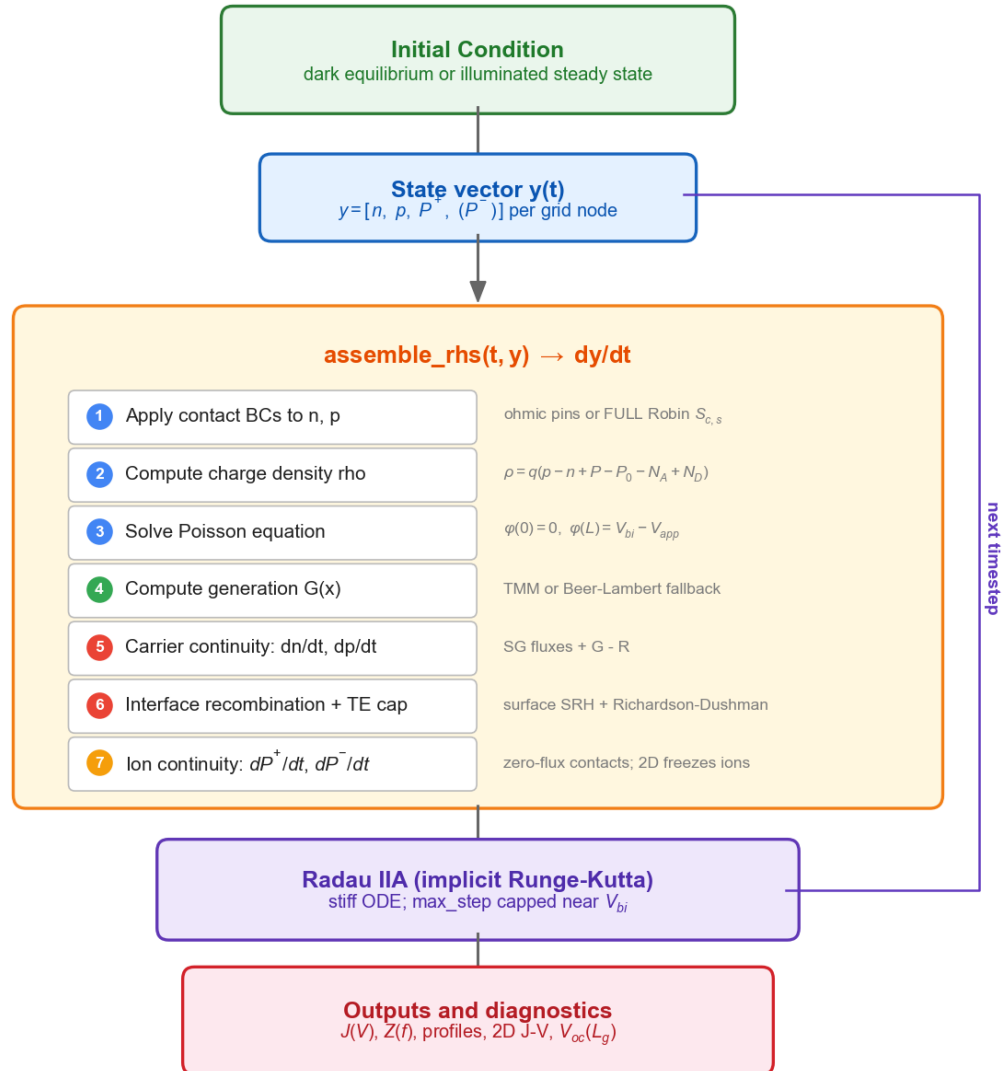


Figure 10.1: SolarLab solver pipeline

10.5 Numerical Safety

Important safety mechanisms include:

- finite-value guards in the RHS;
- harmonic face permittivity;
- stable Bernoulli-function branches;
- TMM transfer-matrix determinant guard;
- median-current steady-state readout;
- explicit `voc_bracketed` flags when a J-V sweep misses the zero crossing.

For publication work, numerical settings are part of the method. Report the grid size, voltage spacing, settling time or scan rate, tolerances when modified, and whether the run used `legacy`, `fast`, or `full` mode.

Chapter 11

Running SolarLab

11.1 Installation

From perovskite-sim/:

```
pip install -e ".[dev]"
```

For the frontend:

```
cd frontend  
npm install
```

11.2 Backend

From the SolarLab root:

```
uvicorn backend.main:app \  
  --host 127.0.0.1 --port 8000 \  
  --app-dir perovskite-sim --reload
```

Do not run uvicorn main:app from inside backend/; the backend imports are written around --app-dir perovskite-sim.

11.3 Frontend

```
cd perovskite-sim/frontend  
npm run dev
```

Then open:

<http://127.0.0.1:5173>

11.4 Python API

```
from perovskite_sim.models.config_loader import load_device_from_yaml  
from perovskite_sim.experiments.jv_sweep import run_jv_sweep
```

```

stack = load_device_from_yaml("configs/nip_MAPbI3.yaml")
result = run_jv_sweep(stack, N_grid=80, n_points=40, v_rate=1.0)

print(result.metrics_fwd.V_oc)
print(result.metrics_fwd.J_sc)
print(result.metrics_fwd.FF)
print(result.metrics_fwd.PCE)

```

11.5 Worked Example 1: Baseline J-V Sweep

This example runs a standard n-i-p MAPbI3 J-V sweep and reports the principal photovoltaic metrics.

```

cd perovskite-sim
python - <<'PY'
from perovskite_sim.models.config_loader import load_device_from_yaml
from perovskite_sim.experiments.jv_sweep import run_jv_sweep

stack = load_device_from_yaml("configs/nip_MAPbI3.yaml")
result = run_jv_sweep(stack, N_grid=80, n_points=40, v_rate=1.0)
m = result.metrics_fwd
print(f"Voc = {m.V_oc:.4f} V")
print(f"Jsc = {m.J_sc:.2f} A/m^2")
print(f"FF = {m.FF:.3f}")
print(f"PCE = {100*m.PCE:.2f} %")
print(f"Voc bracketed: {m.voc_bracketed}")
PY

```

Metric interpretation:

- J_{sc} is the current density near zero applied voltage.
- V_{oc} is interpolated where the terminal current crosses zero.
- FF and PCE are meaningful only when voc_bracketed is true.
- Forward and reverse metrics can differ when mobile ions retain scan history.

11.6 Worked Example 2: Absorber Thickness Study

This example varies absorber thickness while leaving the rest of the stack unchanged, allowing optical collection and recombination trends to be compared under controlled assumptions.

```

from dataclasses import replace
from perovskite_sim.models.config_loader import load_device_from_yaml
from perovskite_sim.experiments.jv_sweep import run_jv_sweep

base = load_device_from_yaml("configs/nip_MAPbI3.yaml")
absorber_index = next(i for i, layer in enumerate(base.layers)
                      if layer.role == "absorber")

for thickness_nm in (200, 400, 700, 1000):
    layers = list(base.layers)
    layers[absorber_index] = replace(

```

```

        layers[absorber_index],
        thickness=thickness_nm * 1e-9,
    )
    stack = replace(base, layers=tuple(layers))
    result = run_jv_sweep(stack, N_grid=80, n_points=30, v_rate=2.0)
    m = result.metrics_fwd
    print(thickness_nm, m.V_oc, m.J_sc, m.FF, m.PCE)

```

Expected interpretation:

- Larger absorbers can absorb more light, but can also increase recombination or transport loss.
- If J_{sc} does not change under Beer-Lambert optics, check whether α and Φ are physically configured.
- If FF collapses, inspect spatial profiles and contact selectivity before concluding that the absorber is intrinsically poor.

11.7 Worked Example 3: Enabling TMM Optics

This example compares scalar Beer-Lambert generation with wavelength-resolved transfer-matrix generation.

```

from perovskite_sim.models.config_loader import load_device_from_yaml
from perovskite_sim.experiments.jv_sweep import run_jv_sweep

for config in ("configs/nip_MAPbI3.yaml", "configs/nip_MAPbI3_tmm.yaml"):
    stack = load_device_from_yaml(config)
    result = run_jv_sweep(stack, N_grid=60, n_points=21)
    print(config, result.metrics_fwd.J_sc)

```

Expected interpretation:

- TMM includes interference, reflection, parasitic absorption, and spectral weighting.
- TMM results are only as credible as the `optical_material` data and AM1.5G spectrum used.
- EQE and EL should use TMM-enabled configurations, not Beer-Lambert-only presets.

Chapter 12

Backend API

Configuration endpoints:

Method	Path	Purpose
GET	/api/configs	list shipped and user presets
GET	/api/configs/{name}	load one preset
POST	/api/configs/user	save a user-edited stack
GET	/api/layer-templates	layer template library
GET	/api/optical-materials	available TMM optical keys

Preferred experiment interface:

Method	Path	Purpose
POST	/api/jobs	submit {kind, config_path device, params}
GET	/api/jobs/{id}/events	stream progress/result/error/done events

Supported job kinds:

jv, current_decomp, spatial, dark_jv, suns_voc, voc_t,
eqe, el, impedance, tpv, degradation, tandem,
mott_schottky, jv_2d, voc_grain_sweep

The frontend primarily uses streaming jobs. Long-running experiments should report progress through callbacks that the backend converts to SSE frames.

Chapter 13

Experiment Manual

13.1 J-V Sweep

The J-V sweep runs forward and reverse voltage scans while preserving state history. This state-preserving scan is the implemented representation of hysteresis in an ion-coupled perovskite cell.

Outputs:

- $V_{\text{fwd}}, J_{\text{fwd}};$
- $V_{\text{rev}}, J_{\text{rev}};$
- $\text{metrics}_{\text{fwd}}, \text{metrics}_{\text{rev}};$
- $\text{hysteresis_index}.$

Metrics:

$$FF = \frac{P_{\text{mpp}}}{V_{\text{oc}} J_{\text{sc}}}.$$

$$PCE = \frac{P_{\text{mpp}}}{1000 W m^{-2}}.$$

If the sweep never crosses $J = 0$, $\text{voc_bracketed}=\text{false}$. In that case V_{oc} , FF, and PCE are sentinel values and the user should increase V_{max} .

13.2 Current Decomposition

Current decomposition separates:

- electron current J_n ;
- hole current J_p ;
- ion current $J_{\text{ion}};$
- displacement current $J_{\text{disp}};$
- total current.

This separation helps identify whether a transient is dominated by electronic, ionic, or capacitive response.

13.3 Spatial Profiles

Spatial-profile runs save snapshots of:

- x ;
- ϕ ;
- E ;
- n ;
- p ;
- P ;
- ρ ;
- applied voltage.

These profiles are essential for interpreting unusual J-V behavior. For example, a high V_{oc} with poor FF may indicate selective-contact or recombination issues rather than poor optical generation.

13.4 Dark J-V

Dark J-V fits the diode equation:

$$J = J_0 \left[\exp \left(\frac{V}{n_{id} V_T} \right) - 1 \right].$$

The output includes ideality factor n_{id} , saturation current J_0 , and the fitted voltage window.

13.5 Suns-Voc

Suns-Voc varies illumination intensity and extracts V_{oc} at each level. It constructs a pseudo-JV curve:

$$J_{\text{pseudo}}(X) = J_{\text{sc,ref}} - J_{\text{sc}}(X), \quad V_{\text{pseudo}}(X) = V_{oc}(X).$$

The pseudo-JV representation is less affected by series resistance than a standard terminal J-V sweep.

13.6 $V_{oc}(T)$

The temperature sweep estimates recombination activation energy by fitting V_{oc} versus temperature:

$$V_{oc}(T) \approx \frac{E_A}{q} - \frac{kT}{q} \ln \left(\frac{J_{00}}{J_{sc}} \right).$$

The intercept at $T = 0$ is reported as E_A in eV.

13.7 EQE / IPCE

EQE is:

$$\text{EQE}(\lambda) = \frac{|J_{\text{sc}}(\lambda)|}{q\Phi_{\text{inc}}(\lambda)}.$$

SolarLab computes monochromatic TMM generation, solves the drift-diffusion problem at short circuit, and integrates against AM1.5G for J_{sc} . This experiment requires `optical_material`.

13.8 Electroluminescence

EL uses reciprocity:

$$\Phi_{\text{EL}}(\lambda) = A_{\text{abs}}(\lambda)\phi_{\text{bb}}(\lambda, T) \exp\left(\frac{qV_{\text{inj}}}{kT}\right).$$

The blackbody photon flux is:

$$\phi_{\text{bb}}(\lambda, T) = \frac{2\pi c/\lambda^4}{\exp(hc/\lambda kT) - 1}.$$

The non-radiative voltage loss is:

$$\Delta V_{\text{nr}} = -V_T \ln(\text{EQE}_{\text{EL}}).$$

This experiment also requires TMM optical data.

13.9 Impedance

Impedance applies a small sinusoidal voltage perturbation around a DC bias and extracts the current response by lock-in fitting. The result is $Z(f)$ in Ωm^2 , suitable for Nyquist and Bode plots.

13.10 Mott-Schottky

Mott-Schottky wraps dark impedance at a fixed frequency and fits:

$$\frac{1}{C^2} = aV + b.$$

Then:

$$V_{\text{bi}} = -\frac{b}{a}, \quad N_{\text{eff}} = -\frac{2}{q\varepsilon_s\varepsilon_0 a}.$$

For thin fully depleted absorbers, this analysis can over-interpret geometric capacitance or nonideal response. Fitted slopes should therefore not be reported automatically as physical dopant densities.

13.11 Transient Photovoltage

TPV applies a small generation pulse at open circuit. SolarLab enforces the open-circuit condition by adjusting V_{app} so that terminal current remains near zero at each reported time. The voltage decay is fitted as:

$$V(t) \approx V_{\text{oc}} + \Delta V_0 \exp(-t/\tau).$$

13.12 Degradation

The degradation experiment is a long-time ion-coupled damage proxy. It advances the ion/carrier system under bias and periodically performs snapshot J-V measurements. During each snapshot, ions are frozen while carriers relax, separating slow ionic redistribution from the instantaneous electronic response.

13.13 Tandem J-V

The tandem driver:

1. runs one combined TMM optical calculation over the full tandem stack;
2. runs independent top and bottom sub-cell J-V sweeps with fixed generation;
3. series-matches at a common current;
4. sums the sub-cell voltages.

Some tandem optical constants are documented as placeholder/stub data. These must be replaced before claiming publication-grade material-specific tandem predictions.

13.14 2D J-V

The 2D J-V driver uses a tensor-product grid. It is validated against 1D in the lateral-uniform limit and can add absorber grain boundaries with reduced SRH lifetimes. Ions are frozen as static Poisson background in 2D.

13.15 $V_{\text{oc}}(L_g)$ Grain Sweep

The grain sweep repeats 2D J-V simulations over grain sizes L_g and reports V_{oc} , J_{sc} , and FF as functions of grain size. It is intended to quantify microstructure-driven voltage loss within the frozen-ion 2D approximation.

Chapter 14

Shipped Presets

Representative shipped presets:

Preset	Purpose
nip_MAPbI3.yaml	n-i-p MAPbI3, Beer-Lambert optics
nip_MAPbI3_tmm.yaml	n-i-p MAPbI3 with TMM optics
pin_MAPbI3.yaml	p-i-n MAPbI3, Beer-Lambert optics
pin_MAPbI3_tmm.yaml	p-i-n MAPbI3 with TMM optics
ionmonger_benchmark.yaml	Courtier/IonMonger-style benchmark
driftdiffusion_benchmark.yaml	Driftdiffusion-inspired benchmark
cigs_baseline.yaml	CIGS-like inorganic thin-film stack
cSi_homojunction.yaml	crystalline silicon homojunction
radiative_limit.yaml	photon-recycling and radiative-limit checks
selective_contacts_demo.yaml	Robin/selective-contact demonstration
field_mobility_demo.yaml	field-dependent mobility demonstration
tandem_lin2019.yaml	2T tandem workflow demonstration
twod/nip_MAPbI3_uniform.yaml	2D lateral-uniform parity preset
twod/nip_MAPbI3_singleGB.yaml	2D single grain-boundary preset
twod/bcx_combined_demo.yaml	combined 2D advanced-physics demo

Preset values are simulation inputs, not universal material constants. For publication studies, users should document the source of every parameter they change or use.

Chapter 15

Troubleshooting And Diagnostics

15.1 `voc_bracketed=false`

Meaning: the J-V sweep did not find a current zero crossing inside the sampled voltage range.

Likely causes:

- V_{max} is too low;
- the voltage grid is too coarse near open circuit;
- the device is numerically unstable near flat band;
- contact or recombination settings produce an unusual current sign.

Recommended response:

1. Increase V_{max} .
2. Increase `n_points` or add a finer voltage region near the expected V_{oc} .
3. Inspect spatial profiles near the highest voltage.
4. Do not report FF or PCE from an unbracketed result.

15.2 Sentinel Or Zero FF/PCE

Zero FF or zero PCE does not always mean a physically dead solar cell. In SolarLab it can also mean the metric extraction refused to infer a maximum power point because V_{oc} was not bracketed. Always check `voc_bracketed` first.

15.3 TMM Or EQE Fails Because Optical Data Are Missing

EQE and EL require at least one layer with `optical_material`. If a stack only uses scalar `alpha`, it can run Beer-Lambert J-V but not wavelength-resolved experiments. Use a `*_tmm.yaml` preset or add verified n, k data.

15.4 YAML Values Look Numeric But Behave Strangely

YAML 1.1 can parse bare scientific notation such as `1e-9` as a string. SolarLab coerces numeric-looking fields in the config loader, but users should write `1.0e-9` for clarity and to avoid surprises in

external tools.

15.5 Slow Or Stiff Runs

Thick absorbers, high fields, sharp interfaces, and ion-coupled transients can make Radau solves slow. Use a coarse grid while developing the configuration, then rerun final cases at publication settings. For CIGS or crystalline silicon, transient J-V may be much slower than equilibrium-style checks.

15.6 Unphysical Band Barriers

Large or inconsistent jumps in χ and E_g can create artificial heterojunction barriers. If a device has unexpectedly poor FF or current, plot the band diagram, check the transport-layer band offsets, and verify the contact velocities.

15.7 2D Results Differ From 1D

For a lateral-uniform 2D stack, differences from 1D should be small when ions are frozen consistently and the grids are comparable. If the difference is large, check:

- whether 1D ions were allowed to move while 2D ions were frozen;
- lateral boundary condition;
- N_x , N_y _per_layer, and settlement time;
- whether a microstructure block was unintentionally active.

15.8 Frontend Job Appears To Hang

Long-running experiments use server-sent events. If the UI does not update, check that the backend was started from the SolarLab root with `--app-dir perovskite-sim`, then verify `/api/jobs/{id}/events` is reachable.

Chapter 16

Validation And Evidence

This manual was generated after a full evidence pass on 2026-05-19.

A Validation evidence executed before manual generation

Gate	Command	Result	Runtime
Default Python suite	pytest	647 passed, 1 skipped	11.7 min
Slow regression suite	pytest -m slow	72 passed	35.9 min
Physics validation suite	pytest -m validation	18 passed	3.6 min
Frontend production build	npm run build	passed	<0.1 min
Frontend unit tests	npm run test:run	320 passed	0.02 min

Figure 16.1: Validation gate summary

Gate	Command	Result	Time
Python default suite	pytest	647 passed, 1 skipped, 72 deselected	704.90 s
Python slow suite	pytest-mslow	72 passed, 648 deselected	2156.96 s
Python validation suite	pytest-mvalidation	18 passed, 702 deselected	217.65 s
Frontend build	npmrunbuild	passed	build step 0.25 s
Frontend unit tests	npmruntest:run	22 files, 320 tests passed	1.29 s

The passing suites cover:

- core Poisson, recombination, ion, optics, contact, temperature, and trap modules;
- J-V, dark J-V, Suns-Voc, EQE, EL, impedance, Mott-Schottky, TPV, degradation, tandem, and 2D workflows;
- backend API and SSE job dispatch;

- frontend result rendering and validation UI;
- IonMonger and Driftdiffusion-inspired benchmark envelopes;
- TMM Jsc baselines;
- photon-recycling voltage boost;
- 1D/2D lateral-uniform parity;
- 2D microstructure regression;
- physical trends for bandgap, thickness, mobility, dark ideality, and Suns-Voc slope.

16.1 Validation Figures

The figures below summarize the validation evidence and slow-suite reference envelopes. They are derived from the tests and evidence files listed in the traceability matrix, rather than from independent simulation runs. The script used to generate them is `docs/manual/generate_manual_figures.py`.

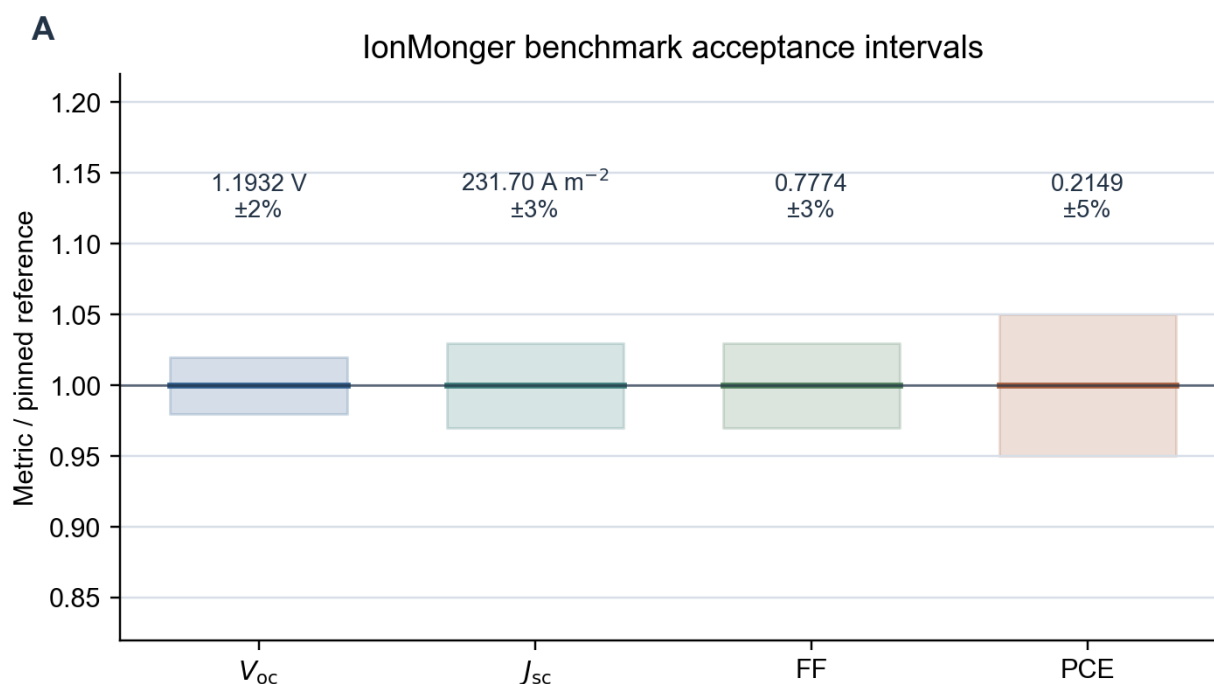


Figure 16.2: IonMonger reference metric envelope

The IonMonger benchmark gate uses `configs/ionmonger_benchmark.yaml` with `N_grid=40`, `n_points=20`, and `v_rate=5.0`. The plotted interval is the allowed tolerance around the pinned reverse-scan reference metrics: $V_{oc} = 1.1932 V$, $J_{sc} = 231.70 A m^{-2}$, $FF = 0.7774$, and $PCE = 0.2149$.

The TMM baseline gate constrains the n-i-p and p-i-n short-circuit currents to remain within the pinned optical-regression tolerance. This protects the transfer-matrix implementation from silent drift while allowing small numerical variation.

The photon-recycling regression requires the radiative-limit V_{oc} boost to remain inside the 40–100 mV literature window encoded by the test. The plotted marker gives the approximate MAPbI3 context value used for the regression.

The validation suite checks qualitative semiconductor-physics trends: current decreases with increas-

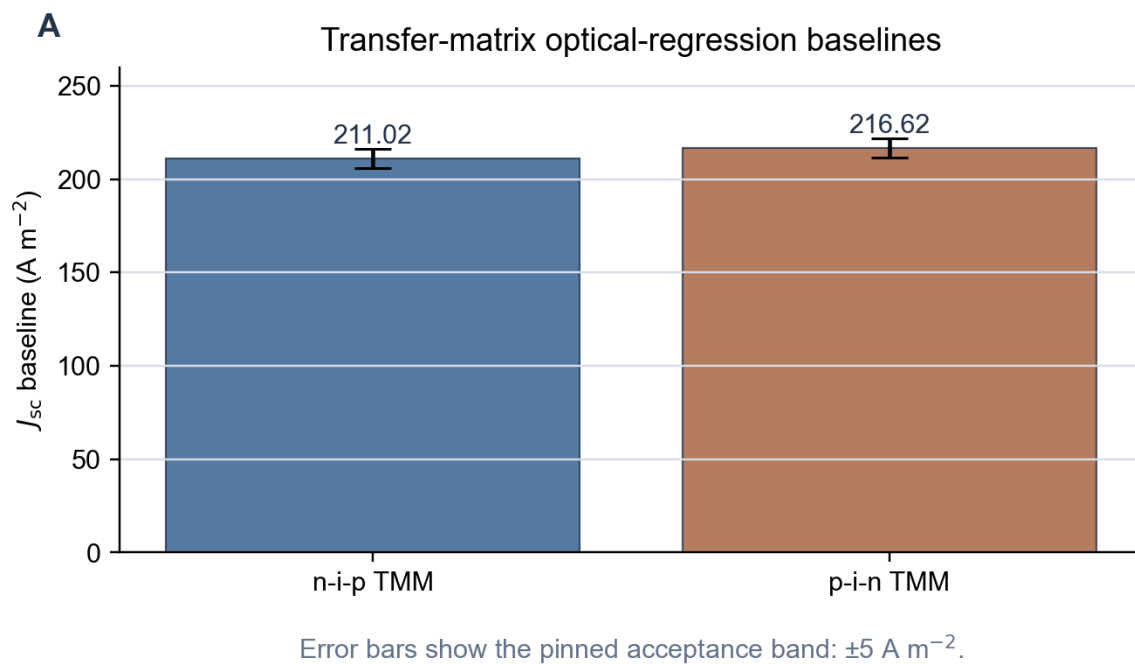


Figure 16.3: TMM optical baseline envelope

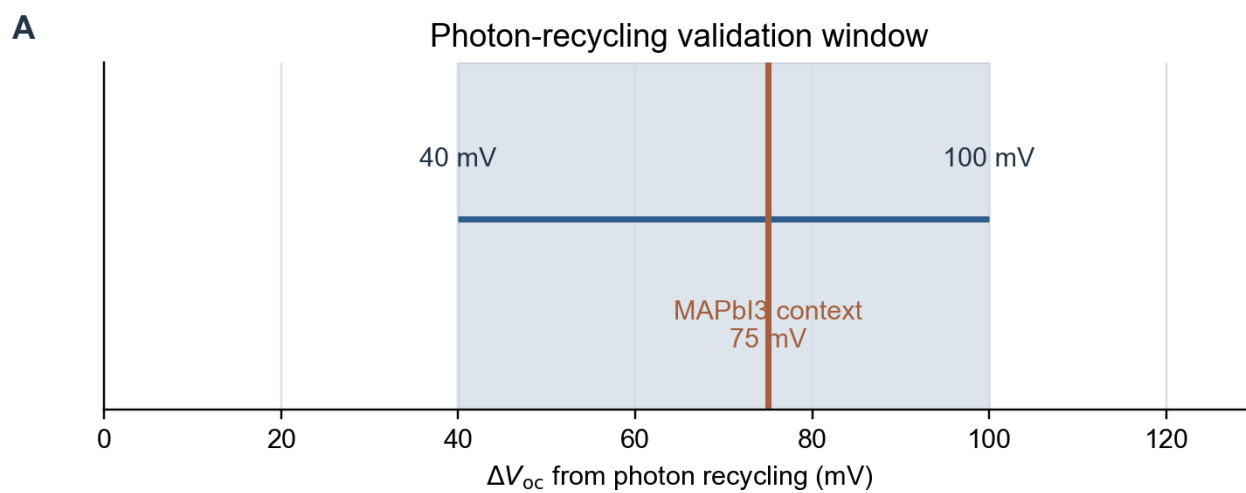


Figure 16.4: Photon-recycling validation window

A

Physics-trend validation matrix

Validation trend	Expected physical behavior	Evidence source
Bandgap	J_{sc} decreases as E_g increases	test_physical_trends.py
Thickness	V_{oc} changes 20-120 mV/decade	test_physical_trends.py
Mobility	FF responds to mobility change	test_physical_trends.py
Dark J-V	$1.0 \leq n_{id} \leq 2.5$	test_physical_trends.py
Suns-Voc	20-70 mV/decade slope	test_physical_trends.py

Figure 16.5: Physics trend validation matrix

ing band gap, mobility affects FF, dark J-V returns an ideality factor in the expected range, and Suns-Voc produces a physically plausible slope.

The 2D slow-suite gates compare the lateral-uniform 2D solver against the 1D reference and verify that a single grain boundary produces a bounded voltage penalty rather than an uncontrolled numerical artifact.

16.2 Validation Traceability Matrix

Evidence item	Source test or file	What it defends
Default Python suite	pytest	broad unit/integration coverage for solver, models, backend, and experiments
Slow Python suite	pytest-mslow	expensive regression baselines, TMM, 2D parity, microstructure, benchmark envelopes
Validation suite	pytest-mvalidation	expected semiconductor-physics trends
IonMonger metric envelope	tests/integration/test_voc_benchmark.py	benchmark-scale J-V metrics and hysteresis bounds
TMM J_{sc} baselines	tests/regression/test_tmm_baseline.py	optical-stack regression stability
Photon-recycling boost	tests/regression/test_photon_recycling_voc.py	radiative-limit voltage boost sanity
2D parity	tests/regression/test_twod_validation.py	lateral-uniform 2D consistency with 1D
Grain-boundary voltage loss	tests/regression/test_twod_microstructure.py	bounded microstructure-induced voltage penalty
Frontend production build	npmrunbuild	TypeScript/Vite production surface
Frontend unit tests	npmruntest:run	UI result rendering, validation, progress, and workstation state

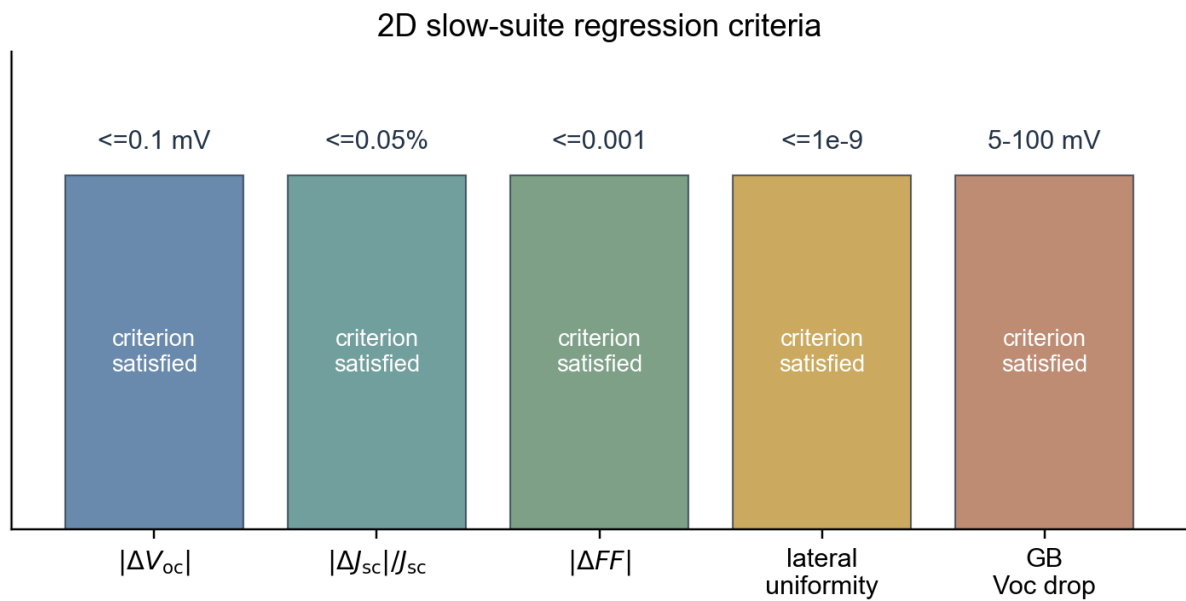
A

Figure 16.6: 2D validation and microstructure summary

Observed warnings:

- `pytest_asyncio` future-default warning for fixture loop scope;
- `np.trapz` deprecation warnings in compatibility and conservation tests;
- Vite bundle chunk-size warning.

None of these warnings are simulator correctness failures, but they should be tracked before a formal software release.

Chapter 17

Limitations And Best Practices

17.1 Model Assumptions

SolarLab is a numerical research simulator. Predictive accuracy for a material stack requires physically justified inputs and a validation envelope that covers the intended regime.

17.2 Data And Optical Limits

TMM, EQE, and EL depend on tabulated n, k data. `load_nk` does not silently extrapolate outside the source wavelength range. Placeholder optical constants support workflow testing, but they do not support material-specific publication claims.

17.3 Numerical Limits

If a J-V sweep does not bracket V_{oc} , increase `V_max` or refine the voltage grid near the expected zero crossing. Sentinel zero FF or PCE should not be interpreted as physical failure until `voc_bracketed` has been checked.

Mott-Schottky fits can be unreliable for fully depleted thin absorbers.

Thick CIGS and crystalline silicon absorbers can be slow in transient mode.

17.4 2D Limits

The 2D solver freezes ions. It is suitable for lateral parity and microstructure studies under that approximation, but not for full dynamic ion migration in two dimensions.

17.5 Reporting Checklist

A reproducible SolarLab study should report:

- commit hash;
- device YAML or full parameter table;
- physics tier;

- grid settings;
- solver tolerances;
- optical data source;
- validation command results;
- whether V_{oc} was bracketed;
- known limitations relevant to the chosen experiment.

17.6 Publication Boundary Statement

A SolarLab figure is publication-ready only when all of the following are true:

- the device stack and all changed parameters are reported;
- optical constants and material parameters have provenance;
- the chosen physics tier is justified;
- the grid and solver settings are sufficient for the conclusion;
- validation gates relevant to the claimed regime have passed;
- limitations such as frozen 2D ions, placeholder optical data, or screening smoke settings are stated explicitly.

If any item is missing, the result may still be useful for debugging or hypothesis screening, but it should not be described as a calibrated material-specific prediction.

Chapter 18

SolarScale Screening Bridge

SolarLab can consume readiness-gated material records from SolarScale. This is a workflow layer, not the core solver.

Safe mappings include:

- dielectric constant to `eps_r`;
- electron and hole mobility to `mu_n` and `mu_p` after unit conversion;
- ion diffusion to `D_ion`;
- ion activation energy to `E_a_ion`.

Device-only unknowns should be swept rather than guessed:

- absorber thickness;
- SRH lifetime;
- trap density;
- interface recombination velocity;
- contact band alignment.

Smoke J-V results in screening workflows validate the process. They are not publication-grade device predictions unless production settings, provenance, and validation are also supplied.

Chapter 19

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Chapter 20

Appendix A: Manual Source Trail

The manual was drafted from:

- docs/solarlab_manual_source_dossier.md
- docs/manual/validation_evidence_2026-05-19.md
- perovskite-sim/AGENTS.md
- README.md
- perovskite-sim/README.md
- perovskite-sim/backend/README.md
- perovskite-sim/perovskite_sim/models/
- perovskite-sim/perovskite_sim/physics/
- perovskite-sim/perovskite_sim/solver/
- perovskite-sim/perovskite_sim/experiments/
- perovskite-sim/perovskite_sim/twod/
- perovskite-sim/backend/main.py
- perovskite-sim/frontend/src/types.ts

This PDF is intended to be defensible as a project manual. For a journal supplement, add parameter provenance for every material value and verify the full bibliography format against the target journal style.

Chapter 21

Appendix B: Figure Source Trail

Figure	Source	Reproducibility note
Architecture flow	docs/manual/generate_manual_figures.py	Diagrammatic summary of repo architecture described in perovskite-sim/AGENTS.md.
Validation gate summary	docs/manual/validation_evidence_2026-05-19.md	Uses pass counts and runtimes from the completed validation pass.
IonMonger reference metrics	tests/integration/test_voc_benchmark.py	Shows pinned benchmark metrics and tolerances, not a new simulation run.
TMM J_{sc} baselines	tests/regression/test_tmm_baseline.py	Shows pinned n-i-p and p-i-n TMM baselines with $\pm 5 \text{ } A \text{ m}^{-2}$ tolerance.
Photon-recycling window	tests/regression/test_photon_recycling_voc.py	Shows the acceptance window used by the slow regression gate.
Physics trend matrix	tests/validation/test_physical_trends.py	Summarizes physical trend assertions represented by the validation suite.
2D validation summary	tests/regression/test_twod_validation.py, tests/regression/test_twod_microstructure.py	Summarizes parity and grain-boundary gates represented by the slow suite.

Chapter 22

Appendix C: Quick Glossary

Term	Meaning
AM1.5G	Standard terrestrial solar spectrum used for photovoltaic testing.
Beer-Lambert	Scalar absorption law using an absorption coefficient α .
Drift-diffusion	Continuum transport model combining diffusion from concentration gradients and drift from electric fields.
FF	Fill factor, the ratio between maximum power and $V_{oc}J_{sc}$.
Hysteresis	Difference between forward and reverse J-V scans caused by state memory, often ionic in perovskites.
Method of lines	Numerical method that discretizes space first and integrates the resulting ODE system in time.
PCE	Power-conversion efficiency.
SG flux	Scharfetter-Gummel flux discretization, used for stable drift-dominated carrier transport.
SRH	Shockley-Read-Hall trap-assisted recombination.
TMM	Transfer-matrix method for wavelength-resolved multilayer optics.
J_{sc}	Short-circuit current density.
V_{oc}	Open-circuit voltage.